

Apurbo Barua

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EDUCATION

University of Arizona

Tucson, AZ

B.S. in Computer Science, Minor in Business Administration | GPA: 3.51/4.00

Aug 2022 – May 2026

- **Honors:** Dean's List, Academic Distinction, Wildcat Excellence Scholarship
- **Leadership:** Chairperson of Academic Excellence, Delta Chi Fraternity

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL

Backend & APIs: Node.js, Express.js, FastAPI, Django, REST APIs, GraphQL, Microservices

Frontend: React, React Native, Expo, Vite, HTML, CSS, Tailwind CSS

Databases & Cloud: PostgreSQL, MongoDB, DynamoDB, Redis, AWS (Lambda, API Gateway, EC2, S3, SAM, IAM)

DevOps & Systems: Docker, Kubernetes, KEDA, Git, GitHub Actions (CI/CD), Linux/Unix, Agile/Scrum, CloudWatch

AI Integration: LLMs, RAG, LangChain, AWS Bedrock, OpenAI/GitHub Models

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Jan 2026 – May 2026

PwC

Tucson, AZ

- Architected a full-stack workforce planning platform (React, TypeScript, Node.js, AWS RDS PostgreSQL) consolidating 5 enterprise systems into a unified dashboard with 5 analytical modules, deployed on AWS EC2.
- Built an AI chatbot using Claude 3 Haiku via AWS Bedrock with a dual-pass engine that generates and executes live SQL against PostgreSQL, enabling stakeholders to query data in plain English.
- Designed and integrated an ML prediction pipeline into the platform's dashboard and REST API layer to surface project delivery risk for staffing decisions.

Software Engineer Intern

Aug 2025 – Dec 2025

EduTrend (Quanticore LLC)

Remote

- Designed GraphQL schemas, resolvers, and MongoDB service layers for admin management in a TypeScript mobile backend.
- Built secure authentication and role-based access control using Clerk, supporting OAuth-based login flows for mobile clients.
- Diagnosed and resolved GitHub Actions CI/CD pipeline failures by fixing cross-platform ESLint and Linux build issues, restoring automated testing and deployment.

Undergraduate Teaching Assistant (CS 110)

Jan 2024 – May 2026

University of Arizona, Dept. of Computer Science

Tucson, AZ

- Built Python automation scripts for grading and testing, reducing manual evaluation time by 10+ hours per week.
- Taught core programming, data structures, and algorithms to 150+ students per semester, debugging Python and Java code in labs and office hours.

PROJECTS

Delta Chi Alumni Portal iamkimball.com

- Architected a centralized alumni networking platform using Django, PostgreSQL, and JavaScript, replacing fragmented Facebook groups to help the Arizona chapter facilitate job sourcing, donations, and newsletters.
- Deployed the application to production on PythonAnywhere, ensuring secure role-based access and reliable uptime for 1,000+ active users.

ResellIQ github.com/ApurboBarua17/ResellIQ

- Architected a resale pricing tool using React, FastAPI, PostgreSQL, and Redis that pulls live eBay and Discogs listings to help resellers price items before listing.
- Integrated GPT-4o-mini to generate recommended asking prices and ready-to-paste listing descriptions from live marketplace data, deployed on a Kubernetes and Docker backend.

AgriSense github.com/ApurboBarua17/AgriSense

- Built a crop-disease detection app using HuggingFace, PyTorch, and GPT-4o-mini that identifies plant diseases from a photo and generates a USDA-backed treatment plan for farmers.
- Deployed 5 FastAPI microservices on Kubernetes with Redis and Meta Prophet to serve real-time disease diagnosis and market price forecasts for crop sell-or-treat decisions.

Website Status Checker github.com/ApurboBarua17/Website_status_checker

- Built a serverless uptime dashboard using React, AWS Lambda, and DynamoDB that executes concurrent Python diagnostic checks to monitor multi-region HTTP and DNS health.
- Provisioned the backend cloud infrastructure with AWS SAM, exposing RESTful APIs to track and report real-time network reliability.

CERTIFICATIONS

IBM Generative AI Engineering – IBM / Coursera

2025

AWS Fundamentals – Amazon Web Services / Coursera

2025